ggplot2 Tutorial for Bioinformatics

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ggplot2 Tutorial for Bioinformatics Analysis

针对生物信息学研究（特别是你正在进行的 RNA-seq 和单细胞分析），ggplot2 不仅仅是一个绘图工具，它是一套基于"图形语法"（Grammar of Graphics）的逻辑系统。

以下是一份为你定制的高度详细教程，涵盖了从基础逻辑到生信实战进阶的内容。

1. ggplot2 的核心哲学：图层 (Layers)

ggplot2 的绘图逻辑就像在透明胶片上叠画：

- **数据层 (Data)**: 你要画什么？
- **映射层 (Aesthetics)**: 数据中的变量对应到画面的什么位置、颜色、形状？========== 最关键！
- **几何对象层 (Geometries)**: 用点、线还是柱状图展示？
- **统计转换层 (Stats)**: 是否需要计算平均值、分位数？
- **标尺控制层 (Scales)**: 颜色用什么色盘？轴的范围是多少？
- **分面层 (Facets)**: 是否要按组分拆成小图？
- **主题层 (Themes)**: 背景、字体、网格线等外观设置。

2. 基础语法模板

```
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.5.2
```

3. 核心要素详解

A. 审美映射 (Aesthetics - aes)

`aes()` 负责将数据列映射到视觉属性。

- `x`, `y`: 坐标轴。
- `color`: 边框或点的颜色。
- `fill`: 填充颜色（如柱状图、箱线图内部）。
- `size`: 大小。

- `shape`: 形状。
- `alpha`: 透明度。

B. 几何对象 (Geoms)
如果你希望表格有标题或引用编号，可以在表格下方紧接着添加：

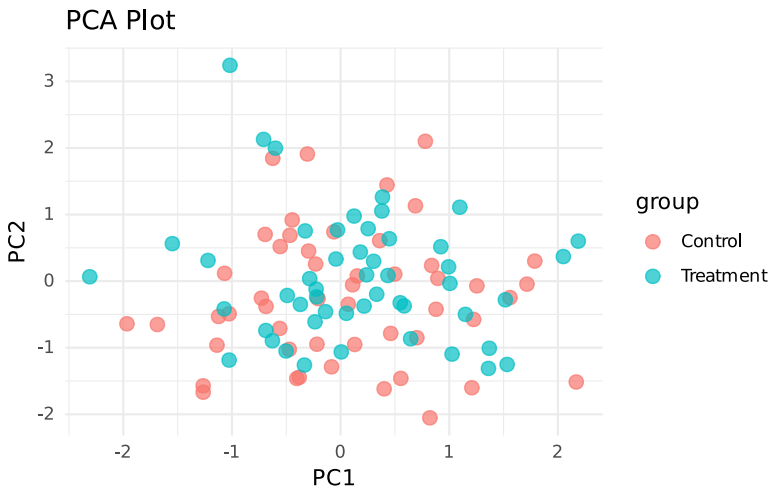
几何对象 (Geom)	数据要求	生信典型应用场景	关键参数建议
`geom_point()`	连续型 x, y	PCA 主成分分析图、火山图 (Volcano Plot)	`alpha=0.5`, `size`
`geom_tile()`	离散型 x, y	差异表达基因热图 (Heatmap)	`aes(fill = value)`
`geom_boxplot()`	离散型 x, 连续型 y	比较不同组别间的基因表达量分布	`outlier.shape = NA`
`geom_violin()`	离散型 x, 连续型 y	单细胞测序中 Cluster 基因分布	`trim = FALSE`
`geom_bar()`	离散型 x	GO/KEGG 富集分析条形图	`stat = "identity"`
`geom_density2d()`	连续型 x, y	**[Wildcard]** 大样本量单细胞轨迹密度分布	`color = "blue"`
`geom_jitter()`	离散型 x, 连续型 y	叠加原始数据点	`width = 0.2`, `alpha = 0.3`

4. Practical Examples

Example 1: Basic Scatter Plot (PCA-like)

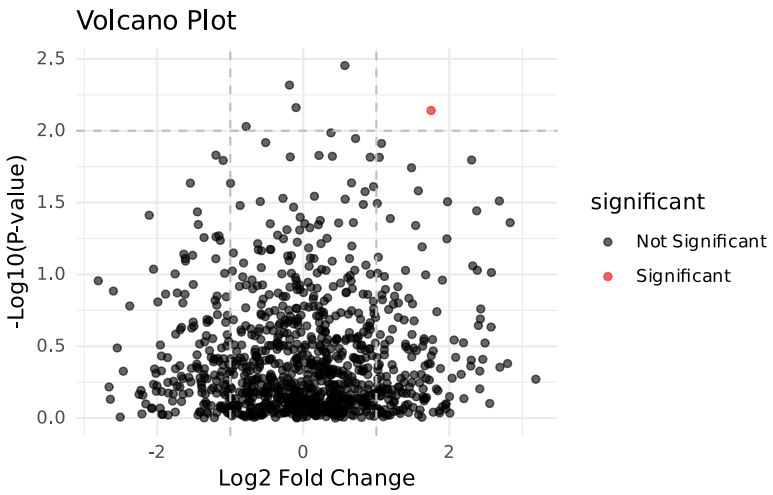
```r
# Load required library
library(ggplot2)

# Create sample data similar to PCA results
set.seed(123)
pca_data <- data.frame(
  PC1 = rnorm(100),
  PC2 = rnorm(100),
  group = factor(rep(c("Control", "Treatment"), each = 50))
)

# Basic scatter plot
ggplot(pca_data, aes(x = PC1, y = PC2, color = group)) +
  geom_point(alpha = 0.7, size = 3) +
```

```
  labs(title = "PCA Plot", x = "PC1", y = "PC2") +
  theme_minimal()
```



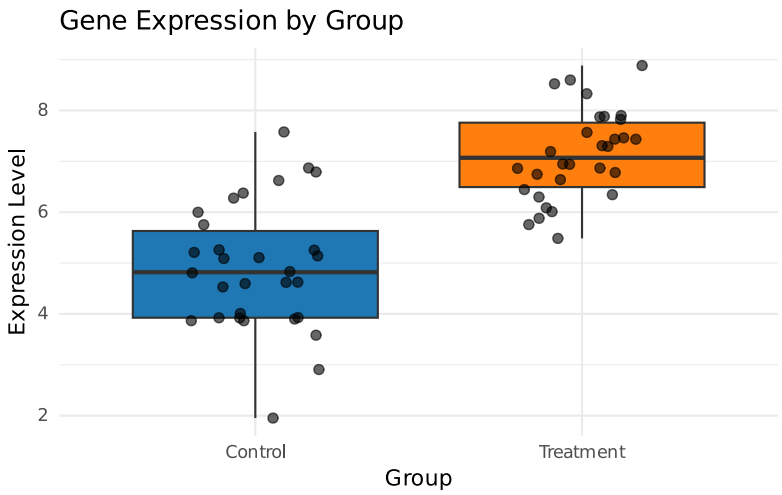
Example 2: Volcano Plot

```
# Create volcano plot data
volcano_data <- data.frame(
  log2FC = rnorm(1000, mean = 0, sd = 1),
  pvalue = runif(1000, min = 1e-10, max = 1)
)
volcano_data$neg_log10_pval <- -log10(volcano_data$pvalue)
volcano_data$significant <- ifelse(volcano_data$neg_log10_pval > 2 &
abs(volcano_data$log2FC) > 1, "Significant", "Not Significant")

# Volcano plot
ggplot(volcano_data, aes(x = log2FC, y = neg_log10_pval, color = significant))
+
  geom_point(alpha = 0.6) +
  scale_color_manual(values = c("Significant" = "red", "Not Significant" =
"black")) +
  labs(title = "Volcano Plot", x = "Log2 Fold Change", y = "-Log10(P-value)")
+
  theme_minimal() +
  geom_vline(xintercept = c(-1, 1), linetype = "dashed", color = "gray") +
  geom_hline(yintercept = 2, linetype = "dashed", color = "gray")
```



Example 3: Box Plot with Jitter Points

```
# Gene expression data across groups
gene_expr <- data.frame(
  expression = c(rnorm(30, mean = 5), rnorm(30, mean = 7)),
  group = factor(rep(c("Control", "Treatment"), each = 30))
)

# Box plot with jitter points
ggplot(gene_expr, aes(x = group, y = expression, fill = group)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.6, size = 2) +
  scale_fill_manual(values = c("Control" = "#1f77b4", "Treatment" =
"#ff7f0e")) +
  labs(title = "Gene Expression by Group", x = "Group", y = "Expression
Level") +
  theme_minimal() +
  theme(legend.position = "none")
```



Introduction to ggplot2

ggplot2 is a powerful and flexible R package for creating elegant data visualizations based on the Grammar of Graphics. This tutorial will guide you through the fundamentals of ggplot2, from basic plots to more advanced customizations.

The Grammar of Graphics

The Grammar of Graphics, developed by Leland Wilkinson, provides a structured approach to describing and building statistical graphics. In ggplot2, every plot is composed of:

1. **Data**: The dataset being visualized
2. **Aesthetics**: How variables map to visual properties (x, y, color, size, etc.)
3. **Geoms**: Geometric objects that represent the data (points, lines, bars, etc.)
4. **Stats**: Statistical transformations (binning, smoothing, etc.)
5. **Scales**: How data values map to visual properties
6. **Coordinate system**: The system used to position elements
7. **Facets**: How to break data into subsets for separate panels
8. **Themes**: Non-data ink (fonts, backgrounds, gridlines, etc.)

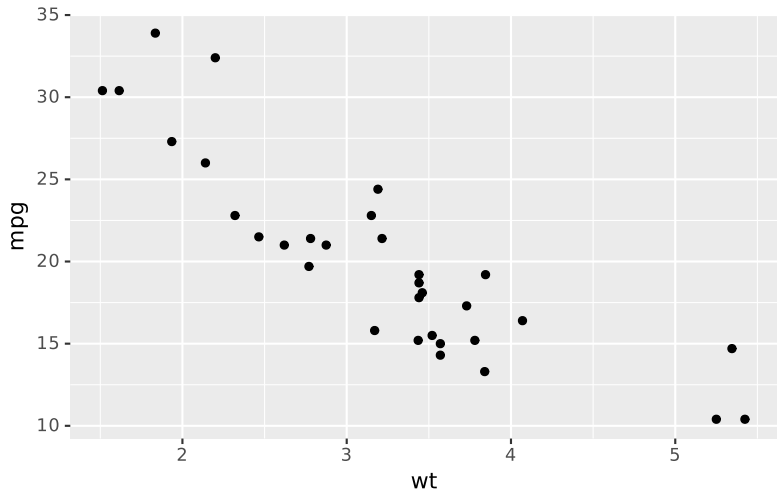
Basic Structure of ggplot2

The basic structure of a ggplot2 plot follows this pattern:

```
# Basic ggplot2 structure
# ggplot(data = <DATA>) +
#   <GEOM_FUNCTION>(
#     mapping = aes(<MAPPINGS>)
#   )
```

Let's start with a simple example using the built-in `mtcars` dataset:

```r
# Simple scatter plot
ggplot(data = mtcars) +
  geom_point(mapping = aes(x = wt, y = mpg))
```



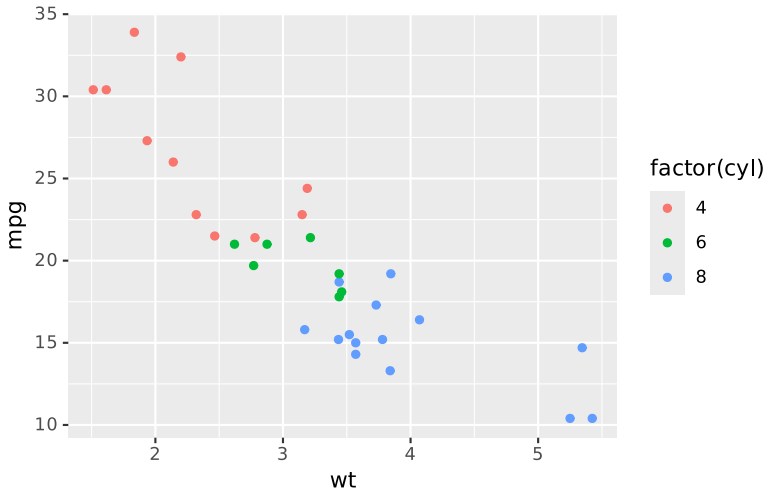
Aesthetics Mapping

Aesthetics define how variables in your data are mapped to visual properties. Common aesthetics include:

- `x` and `y`: Position on x and y axes
- `color`/`colour`: Color of points, lines, or areas
- `fill`: Fill color for areas
- `size`: Size of points or areas
- `alpha`: Transparency (0 = fully transparent, 1 = fully opaque)
- `shape`: Shape of points
- `linetype`: Type of line (solid, dashed, etc.)

Example: Adding Color Based on Cylinders

```r
# Scatter plot with color mapped to number of cylinders
ggplot(data = mtcars) +
  geom_point(mapping = aes(x = wt, y = mpg, color = factor(cyl)))
```



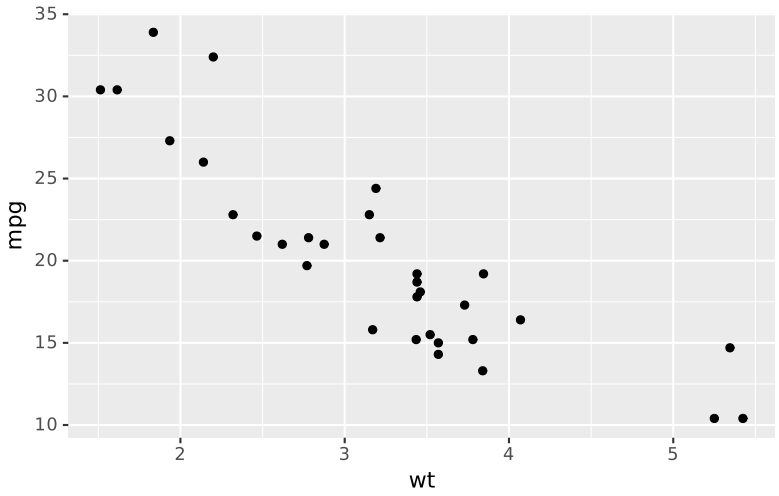
Note that we use `factor(cyl)` to treat cylinder count as a categorical variable rather than continuous.

Common Geoms

Here are some of the most commonly used geometric objects:

Points (`geom_point`)

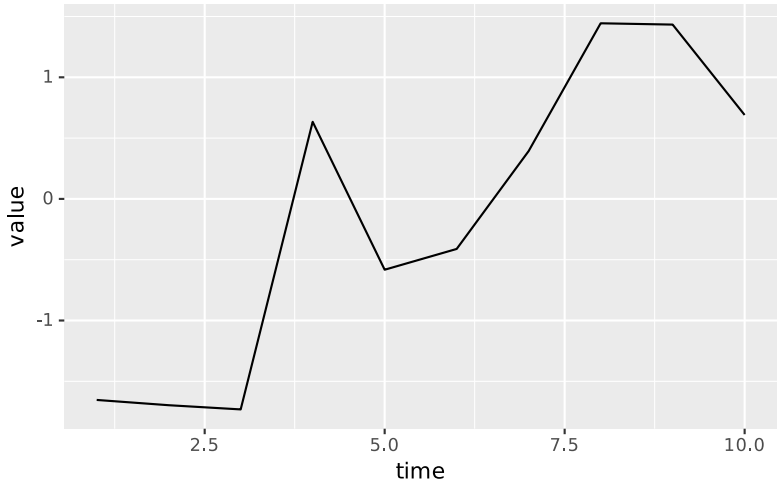
```
ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_point()
```



Lines (`geom_line`)

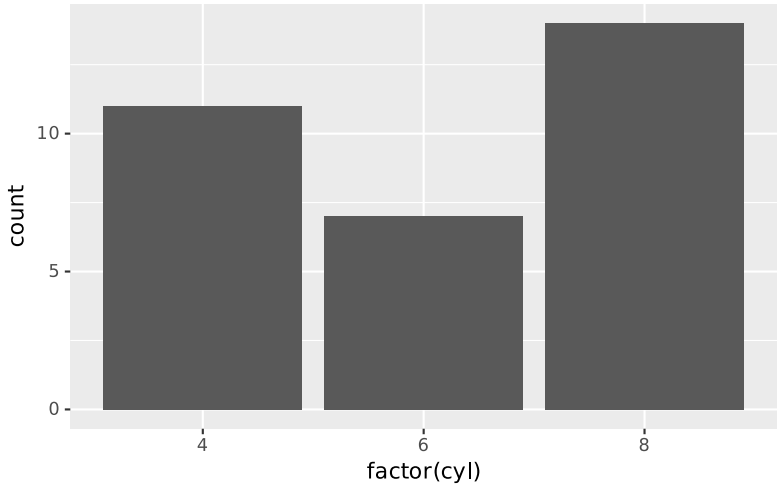
```
# Create time series data
time_data <- data.frame(
  time = 1:10,
```

```
  value = cumsum(rnorm(10))
)

ggplot(time_data, aes(x = time, y = value)) +
  geom_line()
```



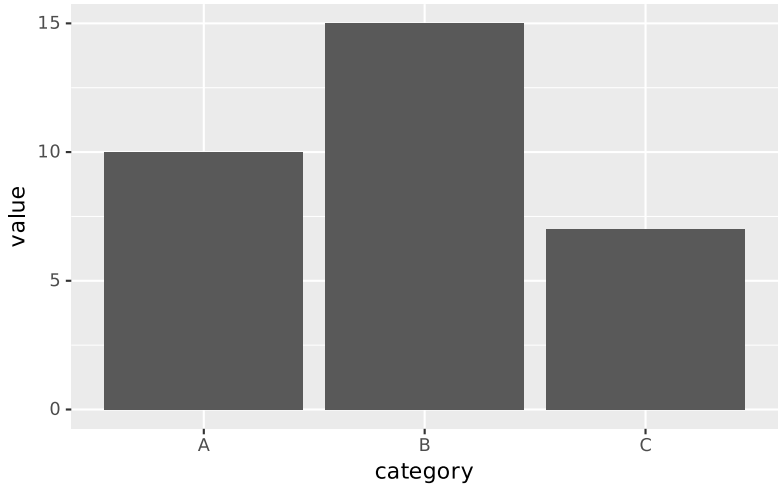
Bars (`geom_bar` and `geom_col`)

```
# geom_bar counts occurrences
ggplot(mtcars, aes(x = factor(cyl))) +
  geom_bar()
```



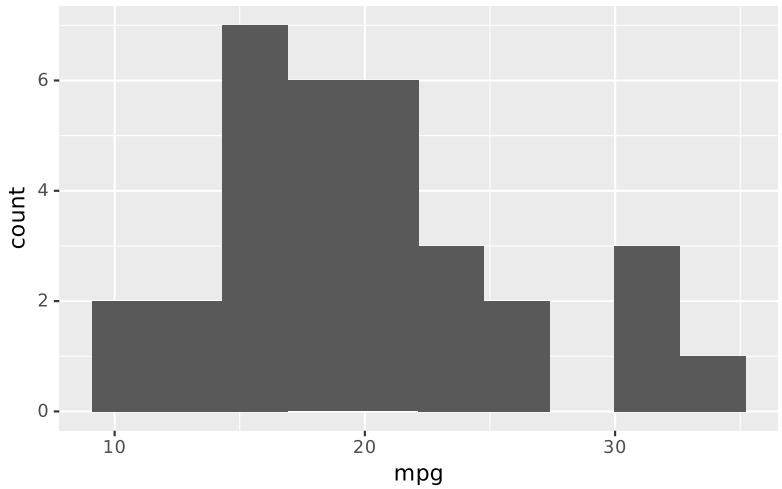

```
# geom_col uses values as heights
bar_data <- data.frame(
```

```
  category = c("A", "B", "C"),
  value = c(10, 15, 7)
)

ggplot(bar_data, aes(x = category, y = value)) +
  geom_col()
```



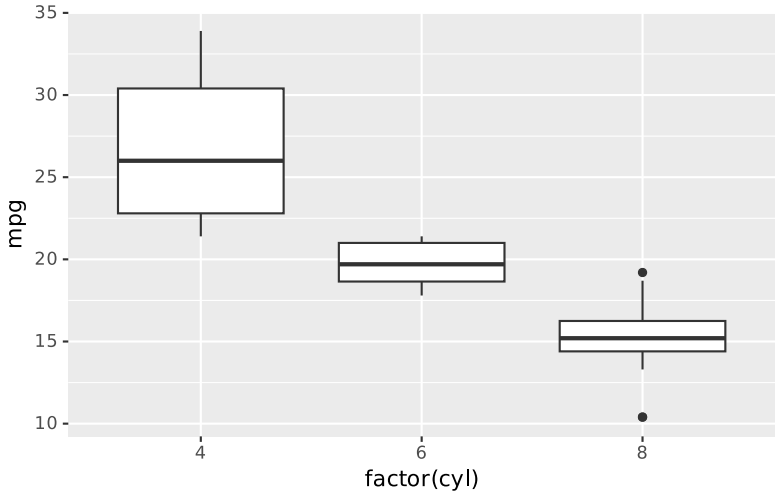
Histograms (`geom_histogram`)

```
ggplot(mtcars, aes(x = mpg)) +
  geom_histogram(bins = 10)
```



Boxplots (`geom_boxplot`)

```
ggplot(mtcars, aes(x = factor(cyl), y = mpg)) +
  geom_boxplot()
```

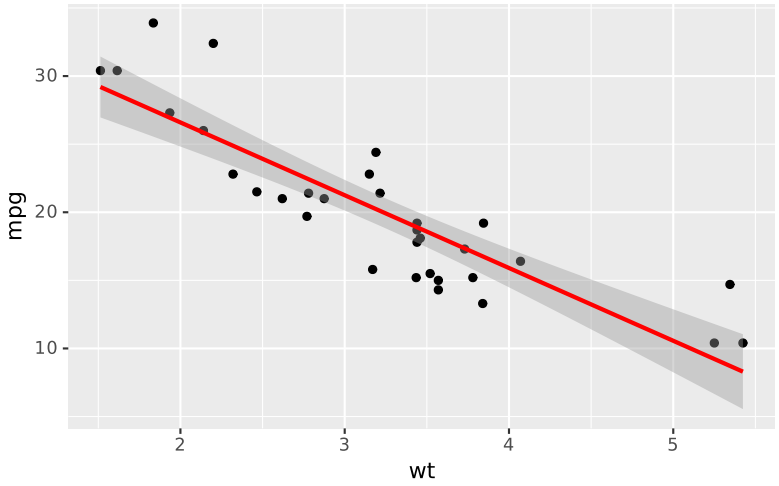


Layers and Multiple Geoms

You can add multiple layers to a single plot:

```
ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_point() +
  geom_smooth(method = "lm", se = TRUE, color = "red")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

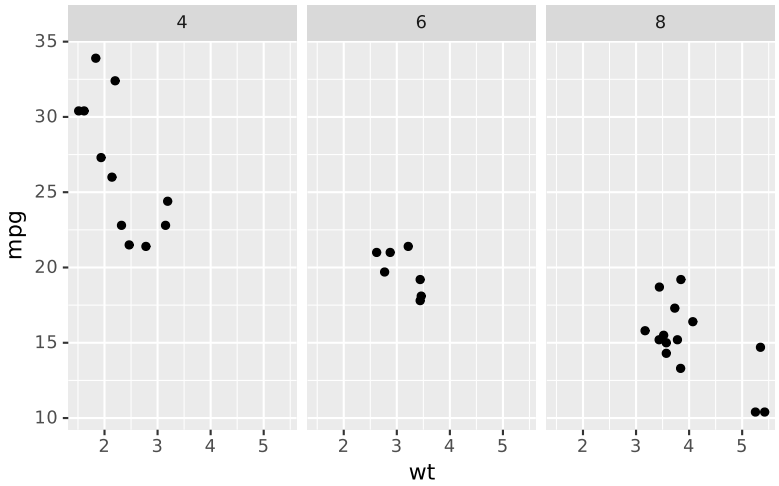


Faceting

Faceting creates multiple plots based on one or more variables:

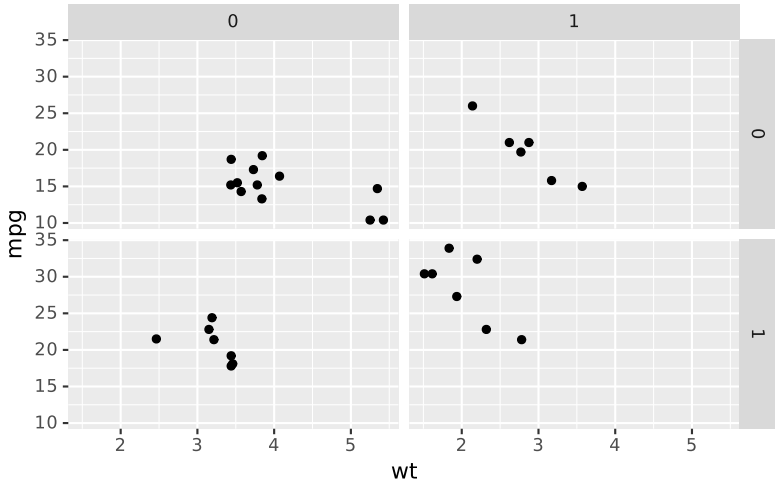
`facet_wrap()`: Wraps facets into a grid

```
ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_point() +
  facet_wrap(~ cyl)
```



`facet_grid()`: Creates a grid of rows and columns

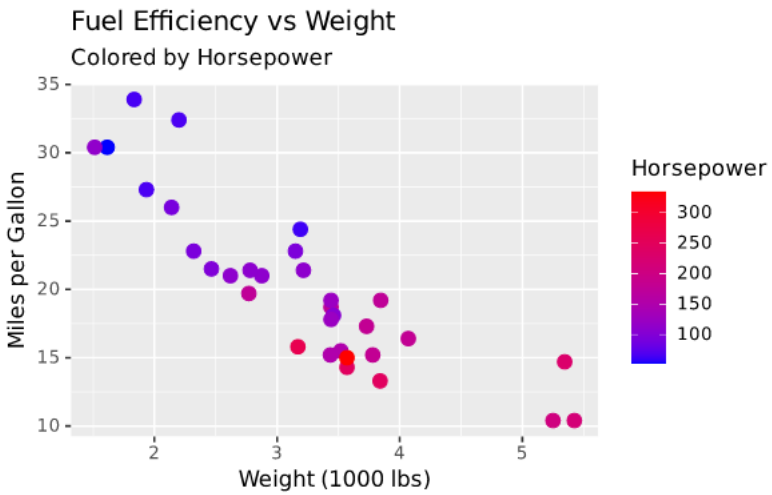
```
ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_point() +
  facet_grid(vs ~ am)
```



Scales and Labels

Customize how data maps to visual properties and add informative labels:

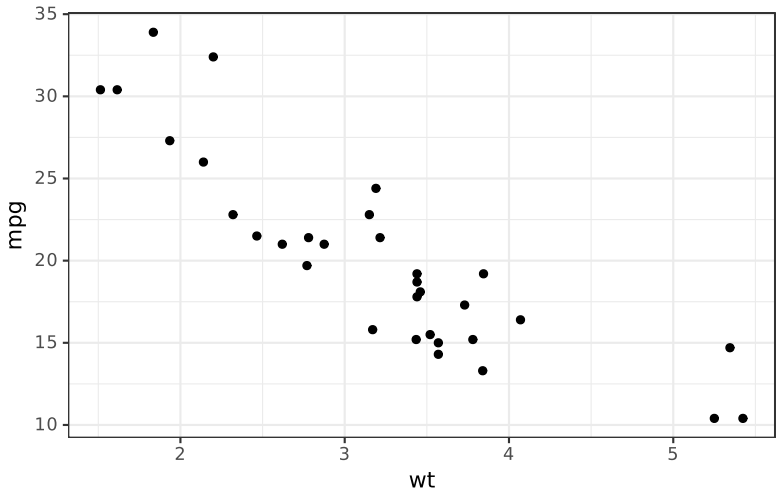
```
ggplot(mtcars, aes(x = wt, y = mpg, color = hp)) +
  geom_point(size = 3) +
  scale_color_gradient(low = "blue", high = "red") +
  labs(
    title = "Fuel Efficiency vs Weight",
    subtitle = "Colored by Horsepower",
    x = "Weight (1000 lbs)",
    y = "Miles per Gallon",
    color = "Horsepower"
  )
```



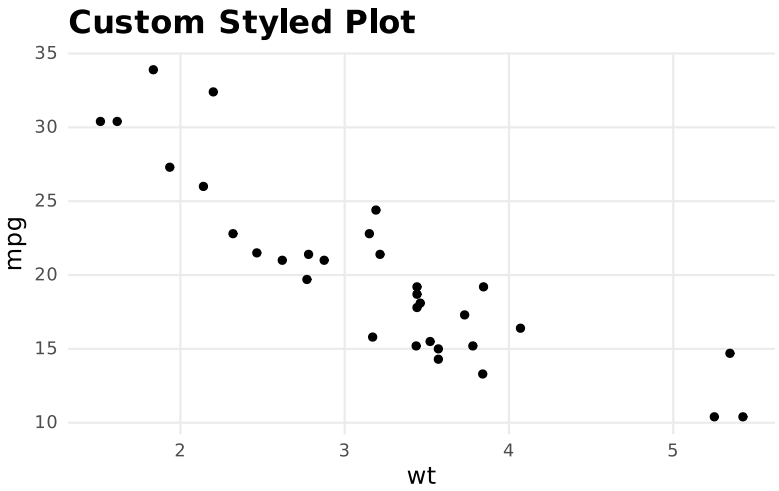
Themes

Themes control the non-data elements of your plot:

```
# Using built-in themes
p <- ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_point()

# Different theme options
p + theme_bw()
```




```
# Custom theme
p + theme_minimal() +
  theme(
    plot.title = element_text(size = 16, face = "bold"),
    axis.title = element_text(size = 12),
    panel.grid.minor = element_blank()
  ) +
  labs(title = "Custom Styled Plot")
```



Saving Plots

Save your plots to files:

```
p <- ggplot(mtcars, aes(x = wt, y = mpg)) + geom_point()
```

```r
# Save as PNG
ggsave("my_plot.png", plot = p, width = 8, height = 6, dpi = 300)

# Save as PDF
ggsave("my_plot.pdf", plot = p, width = 8, height = 6)
```

Best Practices

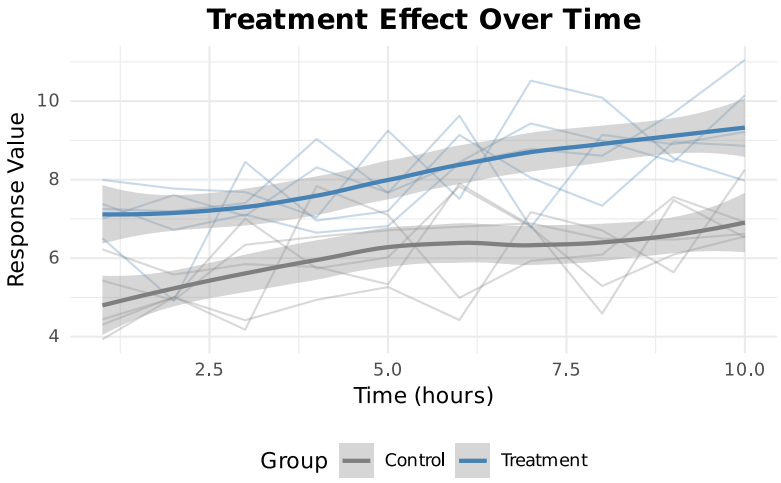
1. **Start simple**: Begin with basic plots and add complexity gradually
2. **Think about your audience**: Choose appropriate colors, labels, and scales
3. **Use meaningful labels**: Always label axes and provide clear titles
4. **Consider colorblindness**: Use color palettes that are accessible to all viewers
5. **Avoid unnecessary elements**: Remove chart junk that doesn't add information

Advanced Example: Multi-panel Publication Quality Plot

Let's create a more complex, publication-ready plot:

```r
# Create sample data
set.seed(123)
sample_data <- data.frame(
  group = rep(c("Control", "Treatment"), each = 50),
  time = rep(1:10, times = 10),
  value = c(rnorm(50, mean = 5), rnorm(50, mean = 7)) +
          rep(seq(0, 2, length.out = 10), times = 10)
)

# Create the plot
ggplot(sample_data, aes(x = time, y = value, color = group)) +
  geom_line(aes(group = interaction(group, rep(1:5, each = 10))),
            alpha = 0.3) +
  geom_smooth(se = TRUE, method = "loess") +
  scale_color_manual(values = c("Control" = "gray50", "Treatment" =
"steelblue")) +
  labs(
    title = "Treatment Effect Over Time",
    x = "Time (hours)",
    y = "Response Value",
    color = "Group"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14, face = "bold"),
    legend.position = "bottom"
  )
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Resources for Further Learning

- ggplot2 official documentation
- R Graph Gallery
- "R Graphics Cookbook" by Winston Chang
- Data Visualization: A Practical Introduction by Kieran Healy

This tutorial covers the fundamentals of ggplot2. With practice, you'll be able to create sophisticated and publication-quality visualizations for your data analysis projects.